

Using sed utility

1. Display the lines that contain the word "lp" in /etc/passwd file.

```
sed -n '/lp/p' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sed -n '/lp/p' /etc/passwd  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin  
_flatpak:x:136:148:Flatpak system-wide installation helper,,,:/nonexistent:/usr/sbin/nologin  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

2. Display /etc/passwd file except the third line.

```
sed '3d' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sed '3d' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
```

3. Display /etc/passwd file except the last line

```
sed '$d' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sed '$d' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
```

4. Display /etc/passwd file except the lines that contain the word “lp”

```
sed '/lp/d' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sed '/lp/d' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
```

5. Substitute all the words that contain “lp” with “mylp” in /etc/passwd file

```
sed 's/lp/mylp/g' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sed 's/lp/mylp/g' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
```

Using AWK utility

1. Print full name (comment) of all users in the system

```
awk -F: '{print $5}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: '{print $5}' /etc/passwd  
root  
daemon  
bin  
sys  
sync  
games  
man  
lp
```

2. Print login, full name (comment) and home directory of all users.(Print each line preceded by a line number)

```
awk -F: '{print NR, $1, $5, $6}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: '{print NR, $1, $5, $6}' /etc/passwd  
1 root root /root  
2 daemon daemon /usr/sbin  
3 bin bin /bin  
4 sys sys /dev  
5 sync sync /bin  
6 games games /usr/games  
7 man man /var/cache/man
```

3. Print login, uid and full name (comment) of those uid is greater than 500

```
awk -F: '$3 > 500 {print $1, $4, $5}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: '$3 == 500 {print $1, $4, $5}' /etc/passwd  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

4. Print login, uid and full name (comment) of those uid is exactly 500

```
awk -F: '$3 == 500 {print $1, $4, $5}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: '$3 > 500 {print $1, $4, $5}' /etc/passwd  
nobody 65534 nobody  
nagy 1000 nagy,,,  
libvirt-qemu 109 Libvirt Qemu,,,  
lxd 135  
user 1000 This is testing useradd cmd  
user2 1000 This is testing useradd cmd  
islam 10000 Islam Asker  
tomcat 10001  
nagy@ZenBook-UX425EA:~$
```

5. Print line from 5 to 15 from /etc/passwd

```
awk -F: 'NR > 5 && NR < 15 {print NR, $1, $4, $5}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: 'NR > 5 && NR < 15 {print NR, $1, $4, $5}' /etc/passwd  
6 games 60 games  
7 man 12 man  
8 lp 7 lp  
9 mail 8 mail  
10 news 9 news  
11 uucp 10 uucp  
12 proxy 13 proxy  
13 www-data 33 www-data  
14 backup 34 backup  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

6. Change lp to mylp

```
awk -F: 'BEGIN{FS=":"} {gsub(/\blp\b/, "mylp"); print}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: 'BEGIN{FS=":"} {gsub(/\blp\b/, "mylp"); print}' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
```

7. Print all information about greatest uid.

```
awk -F: 'BEGIN{max=0} {if($3>max){max=$3; line=$0}} END{print line}'  
/etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: 'BEGIN{max=0} {if($3>max){max=$3; line=$0}} END{print line}' /etc/passwd  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

8. Get the sum of all accounts id's.

```
awk -F: '{sum+=$3} END{print "Sum of all UIDs =", sum}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: '{sum+=$3} END{print "Sum of all UIDs =", sum}' /etc/passwd  
Sum of all UIDs = 176198  
nagy@ZenBook-UX425EA:~$
```

9. Get the sum of accounts id's that has the same group

```
awk -F: '{sum[$4]+=$3} END{for(g in sum) print "gid:", g, " => sum: ", sum[g]}' /etc/passwd
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ awk -F: '{sum[$4]+=$3} END{for(g in sum) print "gid:", g, " => sum: ", sum[g]}' /etc/passwd  
gid: 0 => sum: 0  
gid: 1 => sum: 1  
gid: 2 => sum: 2  
gid: 3 => sum: 3  
gid: 7 => sum: 134  
gid: 8 => sum: 8  
gid: 9 => sum: 9
```